Milestone Three – State-Level Demographic and Socioeconomic Factors Associated With Diabetes Prevalence in the United States

Jonaire Tate

Department of Computing and Data Sciences, Boston University

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Dr. Joshua Von Korff

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1. Description of Project

This project examines whether state-level age distribution and socioeconomic factors are associated with diabetes prevalence in the United States. Specifically, I analyze whether the proportion of older adults (ages 65-74 and 75+), poverty rates, and income per capita are related to state-level diabetes rates. Because diabetes is a chronic condition influenced by both biological and structural factors, examining demographic and economic differences across states may provide insight into broader patterns of health disparities.

The primary research question guiding this analysis is: Are higher proportions of older adults and higher poverty rates associated with higher diabetes prevalence at the state level? Additionally, I examine whether income per capita shows a negative association with diabetes prevalence. This exploratory analysis uses publicly available state-level datasets to investigate these relationships.

2. What I Learned From Exploratory Analysis: Preprocessing

The preprocessing stage required evaluating the suitability of multiple datasets for state-level analysis. Initially, a diabetes dataset obtained from Kaggle was considered; however, it lacked a state identifier column, making it incompatible with the census demographic dataset. Because reliable merging requires consistent geographic keys, this dataset was excluded from further analysis.

A replacement diabetes dataset from the Kaiser Family Foundation (KFF) was selected because it included clearly defined state-level identifiers and age-stratified diabetes prevalence measures. The census demographic dataset was cleaned to retain only state-level aggregates. Column names were standardized across datasets to ensure compatibility. The national aggregate row representing the "United States" was removed to prevent distortion of state-level comparisons.

After cleaning, the datasets were merged using the state variable as the key identifier. The merged dataset was reviewed to confirm that all states were retained and that no unintended missing values were introduced during the merge process. These preprocessing decisions ensured structural consistency and analytical validity prior to exploratory analysis.

Puerto Rico was excluded from the final dataset to maintain consistency with a 50-state analytical framework and to prevent distortion caused by substantially different socioeconomic conditions. The final dataset includes 50 states and the District of Columbia, resulting in 51 geographic observations for analysis.

3. What I Learned From Exploratory Analysis: Univariate Analysis

Diabetes Rate

Figure 1 shows the distribution of state-level diabetes prevalence shows moderate variation, with most states clustered between approximately 7% and 10%. A small number of states exhibit noticeably higher prevalence, creating a mild right tail in the distribution. While extreme outliers are limited, the presence of higher-prevalence states suggests meaningful geographic variation in chronic disease burden.

Income Per Capita

As shown by Figure 2, income per capita demonstrates mild right-skewness, with most states clustered between approximately $20,000 and $35,000. A few higher-income states extend the upper tail of the distribution. This variation reflects economic inequality across states and suggests potential structural differences that may relate to health outcomes.

Poverty Rate

The histogram in Figure 3 visualizes that the poverty rate distribution is moderately right-skewed. Most states fall within the 10%-20% range; however, one state displays a substantially higher poverty rate, creating a visible outlier in the upper tail of the distribution. This skewness indicates that while poverty levels are relatively similar across many states, economic hardship is more concentrated in certain regions.

4. What I Learned From Exploratory Analysis: Bivariate Analysis

Bivariate analysis was conducted to examine relationships between diabetes rates and selected demographic and socioeconomic variables. A strong positive relationship was observed between poverty rate and diabetes prevalence ($r = 0.79$), indicating that states with higher poverty levels tend to experience higher diabetes rates. Income per capita demonstrated a moderate negative correlation with diabetes rate ($r = -0.51$), suggesting that wealthier states generally have lower diabetes prevalence. Additionally, the proportion of residents aged 65–74 showed a clear positive association with diabetes rates, consistent with established age-related risk patterns. Total population size showed minimal correlation ($r = 0.18$), indicating that

overall population volume does not substantially influence diabetes prevalence at the state level.

The combined analysis suggests that socioeconomic conditions are more strongly associated with state-level diabetes prevalence than demographic age structure alone. Poverty rate demonstrated the strongest correlation with diabetes prevalence ($r = 0.79$), indicating a substantial positive association between economic disadvantage and disease burden. Income per capita showed a moderate negative correlation ($r = -0.51$), reinforcing the inverse relationship between economic resources and diabetes rates. While the proportion of residents aged 65–74 was positively associated with diabetes prevalence, its relationship was weaker than that observed for poverty. Figures 4 through 7 provide visualizations of these findings.

These findings suggest that variation in diabetes prevalence across states may reflect structural socioeconomic differences rather than demographic composition alone. Economic disadvantage may influence access to preventive care, nutrition quality, healthcare utilization, and chronic disease management. However, because this analysis is cross-sectional and observational, causal conclusions cannot be drawn. Further multivariate modeling would be required to evaluate independent effects and potential confounding.

5. Conclusions About Data: Expected or Unexpected?

The observed relationships were largely consistent with expectations based on public health literature. Diabetes prevalence was strongly associated with socioeconomic indicators, particularly poverty rate ($r = 0.79$), and moderately negatively associated with income per capita

(r = -0.51). These findings align with established evidence linking economic disadvantage to increased chronic disease burden.

The positive association between the proportion of residents aged 65–74 and diabetes prevalence was also anticipated, as diabetes risk increases with age. No major unexpected patterns were identified in the analysis. The magnitude of the poverty–diabetes correlation, however, was stronger than initially anticipated, suggesting that structural socioeconomic factors may play a particularly influential role at the state level.

6. Conclusions About Data: Datasets Disqualified?

The initial project scope included additional health-related datasets, including maternal and child health indicators and CDC mortality data. These datasets were ultimately excluded because they were not necessary to address the primary research question and would have expanded the scope beyond the intended state-level diabetes analysis.

Additionally, an initial diabetes dataset obtained from Kaggle was excluded because it lacked a state-level identifier, preventing reliable merging with U.S. Census demographic data. To ensure structural consistency and analytical integrity, the Kaiser Family Foundation (KFF) dataset was selected instead, as it provided clearly defined state-level diabetes prevalence measures compatible with census indicators.

7. Conclusions About Data: Supervised or Unsupervised Analysis Suggested?

Given that diabetes prevalence was treated as a defined outcome variable, future analysis would most appropriately employ supervised learning methods. Multiple linear regression could

be used to evaluate the independent contribution of socioeconomic and demographic predictors while controlling for potential confounding variables.

Alternatively, unsupervised clustering techniques could be applied to group states based on similar socioeconomic and demographic profiles, which may reveal structural patterns not captured by correlation analysis alone. However, for predictive or explanatory modeling of diabetes prevalence, supervised approaches would be most appropriate.

8. Figures

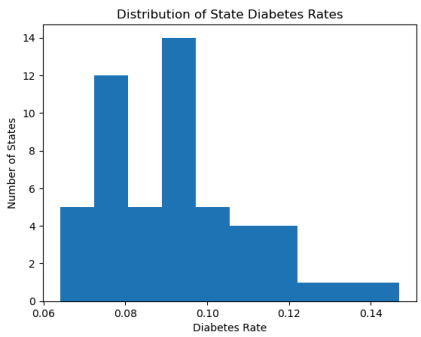

Figure 1. Diabetes Histogram

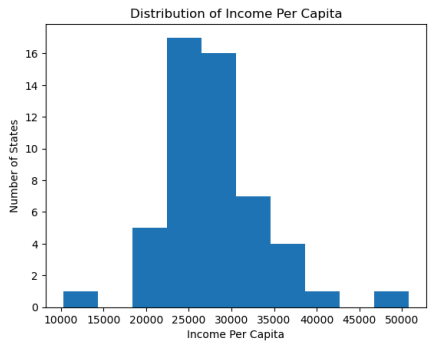

Figure 2. Income Histogram

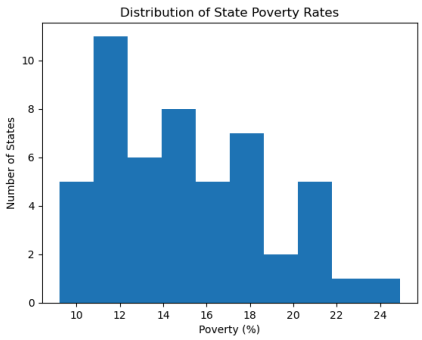

Figure 3. Poverty Histogram

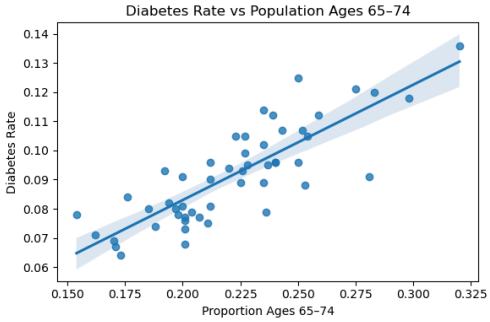

Figure 4. Age vs Diabetes

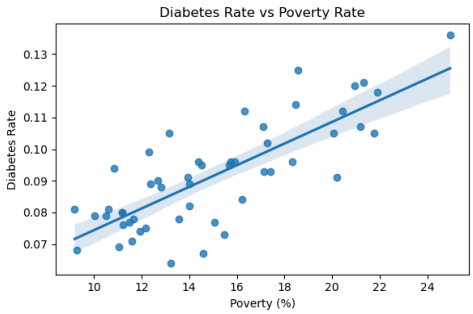

Figure 5. Poverty vs Diabetes

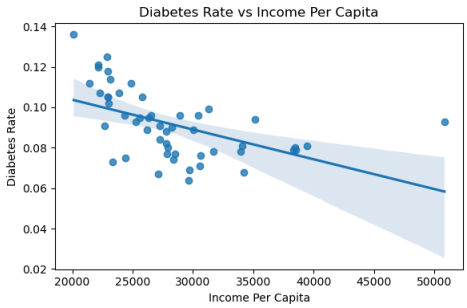

Figure 6. Income vs Diabetes

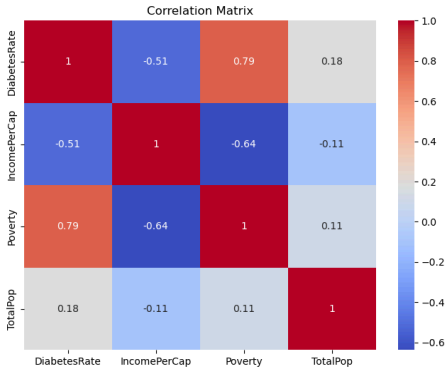

Figure 7. Correlation Matrix

9. References

Kaiser Family Foundation. (2023). *Percentage of Adults with Diagnosed Diabetes by State* [Data set]. Retrieved February 27, 2026, from https://www.kff.org/state-health-policy-data/state-indicator/adults-with-diabetes-by-age/?currentTimeframe=7&sortModel=%7B%22colId%22:%22Location%22,%22sort%22:%22asc%22%7D

Knaflic, C. N. (2015). *Storytelling with Data.* Wiley

U.S. Census Bureau. (2023). *State Demographic and Economic Data (2020-2023)* [Data set]. https://www.kaggle.com/datasets/muonneutrino/us-census-demographic-data?resource=download

10. Appendix A: Weekly Graphs and Homework

- **OMDS-ModB2-Week1-Tate-Jonaire**
- **OMDS-ModB2-Week2-Tate-Jonaire**
- **OMDS-ModB2-Week3-Tate-Jonaire**
- **OMDS-ModB2-Week4-Tate-Jonaire**
- **OMDS-ModB2-Week5-Tate-Jonaire**
- **OMDS-ModB2-Week6-Tate-Jonaire**
- **OMDS-ModB2-Week7-Tate-Jonaire**

11. Appendix B: AI Appendix

Artificial intelligence tools were used to assist with code refinement, statistical interpretation, and structural editing of the written report. All analytical decisions, dataset selection, modeling choices, and interpretations were determined by the author. AI tools were used solely to improve clarity, organization, and technical precision.